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Section- KRG 2B 30/1/26

Competitive Coding

Q⇒ Given three integer n, a, b, return n^{th} magical number. Ans may vary large return 10^9+7

Magical No. ⇒ If Number either divisible by a & b

Approach

Brute ① Take input n, a, b
② Initialize count = 0, i = 1
③ Run a loop:

```
while (count<n):
  • if (i%a==0 || i%b=0)
        count++;
  • i++
```

④ return i-1;

Time Complexity = $O(n*\min(a,b))$ SC = O(1)

Optimal ⇒
① Take input n, a, b
② Calculate LCM,
LCM = (a*b)/gcd(a,b)
③ Set high = min(a*n, b*n);
④ Set low = 0, total = -1
⑤ Run a loop while (low < high)
→ Calculate mid = (low+high)/2;
→ Store three variables:
int x = mid/a
int y = mid/b
int z = mid/lcm

→ Calculate total = x + y − z
→ if (total == n) break;
if (total < n) low = mid+1;
else high = mid;

(6) Return low.

Complexity → "T" O(log n) SC ⇒ O(1)

Code → Bruteforce

```
#include <iostream>
using namespace std;
int magical(int n, int a, int b){
    int i=1, count=0;
    while (true){
        if (i%a==0 || i%b==0){
            count++;
        }
        if (count==n)
            return i;
        i++;
    }
}
int main(){
    int a, b, n;
    cin>>n;
    cin>>a>>b;
int ans = magical(n, a, b);
    return ans;
}
```